

to keep saving the project whenever you make significant changes as you go along.

Now, you can see that there are many things going on here. The spacecraft has to lift off from the launch pad, and make its way over to the landing pad. The controls are space for firing the boosters, A and D keys to rotate the spaceship. There is also a health bar, which starts at 3 and reduces by one every time you impact against an asteroid or another platform. There is also a score indicator that shows if you have successfully executed the task. If you manage to manoeuvre the spacecraft accurately, the UI also displays a congratulatory message.

Now, let us look at what is happening behind the Scenes here. In the Hierarchy panel (on the top left), go to Spaceship. In the Inspector panel (right), you will see a “push strength”, an Axis and a key. The push strength is the amount of movement induced. The axis is the direction of movement in relation to the object. The input key is the button pressed that will trigger the force imparted to the player object, which in this case is the Spaceship. The Spaceship also has a rotation script added, which causes the WASD keys to rotate the spacecraft. The Push and Rotate are the two components, when used in tandem on the Spaceship, cause it to move. The scripts in Unity Playground are all designed to be used with each other in various permutations and combinations, depending on the requirements of the game. The Rigidbody script is there to detect the impact events and reduce the score, but we will come back to that later. What has happened here is that a Ship Body, and two identical Flame gifs were combined into the single Spaceship. This Spaceship was then given the rotate and push scripts, to behave in a certain way depending on the user inputs. You can change some of the values here to make the ship move slower or faster, and this does not have to be just the push strength value. For example, you can reduce the mass of the ship, or reduce the friction, under the Rigidbody component,



The scripts associated with the Spaceship. Many of these values can be changed.